

LDR_LANDER

About :

This feature allows the drone to automatically land when low light is detected, using an LDR (Light Dependent Resistor) sensor.

Add-On used :

An LDR (Light Dependent Resistor) is a sensor that detects light intensity. Its resistance changes depending on how much light falls on it.

More light → less resistance

Less light → more resistance



Components and Connection:

Pluto drone, LDR sensor module and connecting wires.

Connect the LDR sensor on the ADC port of Primus V5/X2 board

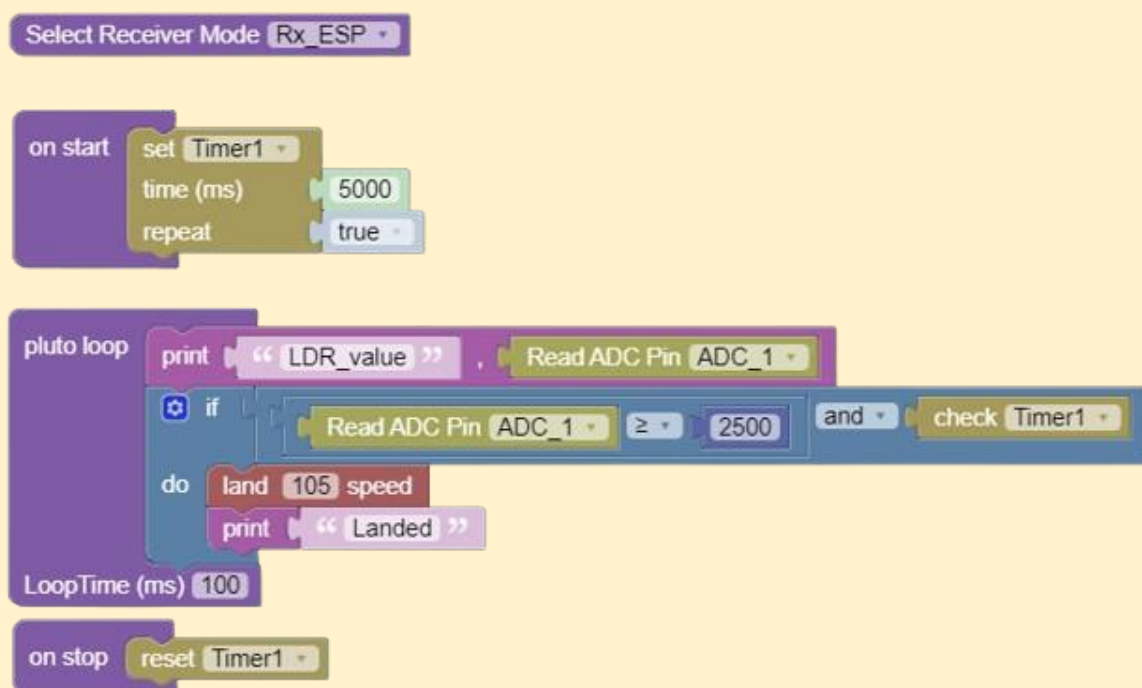
Primus V5/X2 pin	Peripherals
VCC on ADC port	VCC on LDR Sensor
GND on ADC port	GND on LDR Sensor
Signal on ADC port	A0 on LDR Sensor

How does it work?

The drone flies normally under user control. When developer mode is enabled and a low-light condition is detected for a fixed time, the drone initiates a safe automatic landing.

When LDR notices any shadow/ lowlight its value goes above 2500. Landing the drone requires approximately 5 seconds. So when timer flag is on and drone is in dark, the drone lands safely before timer flag turns on again.

PlutoBlocks



Working:

Initially, we set Receiver mode as ESP

Then, we set the timer for 5000 milli second(5 sec). If the sensor detects value more than 2500 it indicates drone has entered a dark area and so upon timer flag turning on, it lands.



fig: LDR addon with Pluto

Moving further

Try making a different use of LDR to make drone behave a differently.

Try making a flash light follower.